

Verification as a Costed Action: When Checking Beats Containing Under Attack–Fault Ambiguity

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Abstract

Draft pending. Claim: verification is worth its cost exactly over an interval of beliefs around the containment threshold—a two-threshold leave/verify/contain rule—and that interval vanishes at a computable critical delay.

1. Introduction

Planned. Opening argument: Paper 1’s identifiability limit bounds what *order-flow* observation reveals; verification reads a separate channel (authorization), which that bound does not cover.

2. Related work

Planned. Costly information acquisition, value of information, active diagnosis, intrusion response.

3. Model

An operator watches an anomaly on a trading gateway and must decide what caused it. Two causes are possible: a *compromise* K , in which an adversary is acting through a legitimate account or session, and a *fault* F , in which the same symptoms come from a benign malfunction. The operator can *contain* (suspend the account or session, pull the orders) or *leave* (let trading continue). Before committing to either, the operator may also *verify*: query a source of evidence that the order flow does not carry. This section sets out the decision problem; Section 4 solves it.

3.1. Loss table and passive risk

The operator's belief that the cause is a compromise is $q \in [0, 1]$. Without verification, the operator picks the terminal action with the lower expected loss.

Definition 1 (Loss table, passive risk). There are two causes, compromise K and fault F , and two terminal actions, *contain* and *leave*. Containing costs C_K under K and C_F under F ; leaving costs H_K under K and H_F under F . For a belief $q = P(K) \in [0, 1]$ the *passive risk* is

$$R(q) = \min\{q C_K + (1 - q) C_F, q H_K + (1 - q) H_F\}.$$

Write $a = H_K - C_K$ and $b = C_F - H_F$.

The four entries of the loss table are kept general. Containing a compromise (C_K) costs the disruption of a legitimate client plus the operational cost of the response; containing a fault (C_F) costs the same disruption with nothing gained; leaving a compromise (H_K) costs the harm the adversary goes on to do; leaving a fault (H_F) costs whatever the malfunction costs if it is not interrupted. The quantity a is what containment saves when the cause is a compromise, and b is what it wastes when the cause is a fault. The cost matrix of Paper 1 is the special case $H_F = 0$, with Paper 1's containment cost, disruption and harm in the roles of C_K , C_F and H_K .

3.2. Verification

Verification is a costed action with an uncertain outcome. On a trading gateway it is, for example, a validation of the credential or session that submitted the orders, or a call from the desk to the client to confirm the activity.

Definition 2 (Verification action). Verification returns an outcome $Z \in \{+, -\}$ with sensitivity $s = P(Z = + | K)$ and specificity $t = P(Z = - | F)$. It costs c_v and takes a delay τ ; during the delay a true compromise causes harm at rate h_K . From belief q ,

$$P(+) = qs + (1 - q)(1 - t), \quad P(-) = q(1 - s) + (1 - q)t,$$

$$q^+ = \frac{qs}{P(+)}, \quad q^- = \frac{q(1 - s)}{P(-)}$$

(a posterior whose outcome has probability zero is never used; see Definition 3).

The sensitivity s is the probability that verification flags a compromise that is present; the specificity t is the probability that it clears a fault. The cost c_v is paid whatever the cause: analyst time, the call, the friction imposed on the client. The delay τ is the time the check takes, and the harm rate h_K is what an adversary who is really present gains per unit of that time. A fault does not grow worse because the operator is checking; a compromise does.

Definition 3 (Verify-first risk, value of information, verification region). The *verify-first risk* is

$$R_v(q) = c_v + q h_K \tau + P(+) R(q^+) + P(-) R(q^-),$$

where $P(z) R(q^z)$ is read as $\min\{ P(z, K) C_K + P(z, F) C_F, P(z, K) H_K + P(z, F) H_F \}$ with $P(z, K) = P(z \mid K) q$ and $P(z, F) = P(z \mid F) (1 - q)$ (equal to the product when $P(z) > 0$, and zero when $P(z) = 0$). The *value of information* is $V(q) = R(q) - [P(+) R(q^+) + P(-) R(q^-)]$, the *verification cost* is $\kappa(q) = c_v + q h_K \tau$, and the *verification region* is

$$S = \{ q \in [0, 1] : R_v(q) < R(q) \} = \{ q \in [0, 1] : V(q) > \kappa(q) \}.$$

Ties are resolved in favour of the passive action: the operator verifies only when verifying is strictly cheaper.

The verify-first risk adds three terms to the expected terminal loss after the update: the cost of the check, the harm expected during the delay, and the expected passive risk at the updated belief. The value of information V is the familiar reduction in terminal risk that the outcome buys [1]; the verification cost κ collects what the check itself costs. Because harm during the delay accrues only under K , κ grows with the belief: checking is more expensive precisely when a compromise is more likely. The verification region S is the set of beliefs at which the check pays for itself.

3.3. Assumptions

Assumption 1 (Cost orderings). $H_K > C_K$ and $C_F > H_F$; that is, $a > 0$ and $b > 0$: leaving a compromise is strictly worse than containing it, and containing a fault is strictly worse than leaving it.

This excludes the uninteresting cases in which one terminal action dominates the other whatever the cause. On a gateway it says that an unchecked adversary does more damage than a response costs, and that suspending

a healthy client costs more than letting a benign fault run. Remark 1 in Section 4 records what happens when it fails.

Assumption 2 (Separate channel). Verification reads authorization (credential or session validity), not order flow. Its outcome Z is conditionally independent of the order-flow observation Y given the cause: $P(Z, Y | c) = P(Z | c) P(Y | c)$ for $c \in \{K, F\}$, and (s, t) do not depend on Y .

Assumption 2 is the conceptual core of the paper and is discussed in Section 3.4.

Assumption 3 (Informative one-shot verification). $s, t \in [0, 1]$ with $s + t > 1$; $c_v \geq 0$, $\tau \geq 0$, $h_K \geq 0$. Verification is available once; after its outcome the operator takes a terminal action (contain or leave) and cannot verify again.

The condition $s + t > 1$ says that a positive outcome is more likely under a compromise than under a fault; a check that fails it carries no information (Remark 1). The one-shot restriction matches a triage step in which one check is run and the operator then acts. Repeated or sequential checking is a stopping problem and is outside the scope of this paper.

Assumption 4 (Delay accounting). The harm $h_K \tau$ accrued by a true compromise during the verification delay is *additional* to the terminal cost (C_K or H_K) incurred afterwards, whichever terminal action is taken; the terminal costs C_K, C_F, H_K, H_F and the verification characteristics s, t do not depend on τ ; a fault accrues no additional cost during the delay; and no further order-flow evidence is used between the start of verification and the terminal action.

Assumption 4 fixes how the delay is charged. The harm accrued during the check is paid in addition to the terminal cost, whatever the operator then does; this is the conservative accounting for verification, since it charges the delay in full even when the terminal cost H_K might already include part of it. The check does not change the loss table or its own accuracy, a fault costs nothing extra while it is checked, and the operator does not update on further order flow while waiting for the outcome.

3.4. Why verification is not bound by the order-flow limit

Paper 1 shows that order-flow observation cannot separate a compromise from a fault when the two causes are confounded, and that a passive operator who sees only order flow then pays a cost of ambiguity that no classifier can remove. That limit is a statement about the law of the order-flow observation

Y under the two causes; it constrains every decision rule that is a function of Y alone. It does not constrain evidence that is not a function of Y .

Verification is such evidence. A credential or session check reads the authorization channel: whether the entity submitting the orders is the one entitled to submit them. Assumption 2 states that, given the cause, the outcome of that check is independent of what the order flow shows, and that its accuracy (s, t) does not depend on the order flow. Operationally, whether a session token is valid does not depend on how the orders it submitted look. Under this assumption verification composes with whatever the passive analysis concludes: Lemma 2 in Section 4 shows that the joint posterior is the verification update applied to the order-flow posterior, with the same characteristics (s, t) whatever the order-flow law. In particular it applies when the order-flow laws under K and F coincide. That is the case in which Paper 1’s passive operator can do no better than act on the prior; Corollary 1 in Section 4 states when verification brings the cost of ambiguity strictly below Paper 1’s value there.

Assumption 2 is a modelling choice, not a consequence of the setting. Without it the conditional law of the check’s outcome given the cause may depend on the order-flow observation, the verification update is no longer a function of the order-flow posterior alone, and Lemma 2 no longer holds.

4. The two-threshold rule and the critical delay

This section derives when verification pays. The passive rule has a single threshold (Lemma 1). Verification composes with any order-flow posterior (Lemma 2), and its value is a tent centred on that threshold (Lemma 3). Comparing the tent with the cost of the check gives a leave / verify / contain rule with two thresholds (Theorem 1), and a critical delay beyond which the check never pays (Theorem 2). Corollary 1 applies these results to the confounded case, and Remark 1 records the degenerate parameters. Proofs are collected in Section 4.8.

4.1. The passive threshold

Lemma 1 (Passive threshold). Under Assumption 1, let $q^* = b/(a + b) \in (0, 1)$. Leaving is strictly optimal for $q < q^*$, containing is strictly optimal for $q > q^*$, and

$$R(q) = q H_K + (1 - q) H_F - (a + b) (q - q^*)^+,$$

so R is continuous, concave and piecewise linear on $[0, 1]$ with a single kink at q^* .

The operator who cannot verify contains exactly when the belief in a compromise exceeds q^* : the share of the total stakes $a + b$ that a wrong containment wastes. A response that is costly to a healthy client (large b) pushes q^* up; an adversary who does much damage if left alone (large a) pulls it down. At q^* the two terminal actions are equally good, and the passive risk has its only kink.

4.2. Verification composes with the passive posterior

Lemma 2 (Verification composes with any passive posterior). Under Assumption 2, let $q(y) = P(K \mid Y = y)$ be the passive (order-flow) posterior. Then $P(K \mid Y = y, Z = z)$ equals the verification update of Definition 2 applied to $q = q(y)$: it is $q(y)^+$ if $z = +$ and $q(y)^-$ if $z = -$. In particular the characteristics (s, t) , and hence V , S and all results below, are the same whatever the order-flow law, including when the order-flow laws under K and F coincide.

Lemma 2 separates the two stages of the decision. The operator first reduces the order-flow evidence to a belief $q(y)$, with whatever classifier or model is available, and then applies the verification analysis below to that belief, with the same (s, t) . The verification rule need not be re-derived for each classifier, and it remains meaningful when the order flow carries no information about the cause (Section 3.4). All results below are stated for a belief q and hold at $q = q(y)$.

4.3. The value of information is a tent

Lemma 3 (The value of information is a tent). Under Assumptions 1 and 3,

$$V(q) = \max\{0, \min\{G^+(q), G^-(q)\}\}, \quad G^+(q) = q s a - (1-q)(1-t)b, \quad G^-(q) = (1-q)t b - q a$$

Consequently, with

$$q_L = \frac{(1-t)b}{s a + (1-t)b}, \quad q_U = \frac{t b}{t b + (1-s)a},$$

(i) $0 \leq q_L < q^* < q_U \leq 1$; (ii) $V(q) > 0$ if and only if $q \in (q_L, q_U)$, equivalently if and only if $q^- < q^* < q^+$ (the two posteriors fall strictly on opposite sides of q^*); (iii) V is affine and strictly increasing on $[q_L, q^*]$, affine and strictly

decreasing on $[q^*, q_U]$, and zero outside (q_L, q_U) ; (iv) V attains its maximum uniquely at q^* , with

$$V(q^*) = (s + t - 1) \frac{a b}{a + b} = (s + t - 1) a q^* > 0.$$

Lemma 3 is the standard value-of-information calculation for two actions and two states [1]. A check is worth something only when its outcome can change the terminal action: a positive outcome must move the belief above q^* and a negative outcome below it. Far from the threshold neither outcome can overturn the passive decision, and the check is worthless however accurate it is. The value rises linearly from q_L to its peak at q^* and falls linearly to q_U . The height of the peak is the product of the Youden index $s + t - 1$ and the quantity $a q^* = b(1 - q^*)$, which is the expected regret at the threshold, relative to an operator who knows the cause, of either terminal action.

4.4. The two-threshold rule

Theorem 1 (Two-threshold leave / verify / contain rule). Under Assumptions 1, 3 and 4, the verification region S is an open interval (possibly empty). It is non-empty if and only if

$$V(q^*) > c_v + q^* h_K \tau,$$

and then $S = (q_1, q_2)$ with $q_L \leq q_1 < q^* < q_2 \leq q_U$, where

$$q_1 = \frac{c_v + (1 - t) b}{s a + (1 - t) b - h_K \tau}, \quad q_2 = \frac{t b - c_v}{t b + (1 - s) a + h_K \tau}$$

(both denominators are positive when $S \neq \emptyset$). The optimal one-shot policy is: *leave* if $q \leq q_1$, *verify* if $q_1 < q < q_2$ and then contain on $Z = +$ and leave on $Z = -$, *contain* if $q \geq q_2$ (with $q_1 = q_2 = q^*$ and the passive rule of Lemma 1 when $S = \emptyset$). When $c_v = 0$ and $\tau h_K = 0$, $S = (q_L, q_U)$.

Theorem 1 is a transposition to attack-fault triage of the threshold approach of **(author?)** [2], in which a clinician's probability of disease is split by a testing threshold and a test-treatment threshold into no treatment, testing and treatment. Here q_1 plays the role of the testing threshold and q_2 that of the test-treatment threshold, with contain in place of treat and leave in place of withhold, and with a general loss table (C_K, C_F, H_K, H_F) . The result is not new as decision theory; what the transposition adds is the cost of the check that grows with the belief, through the delay term $q h_K \tau$.

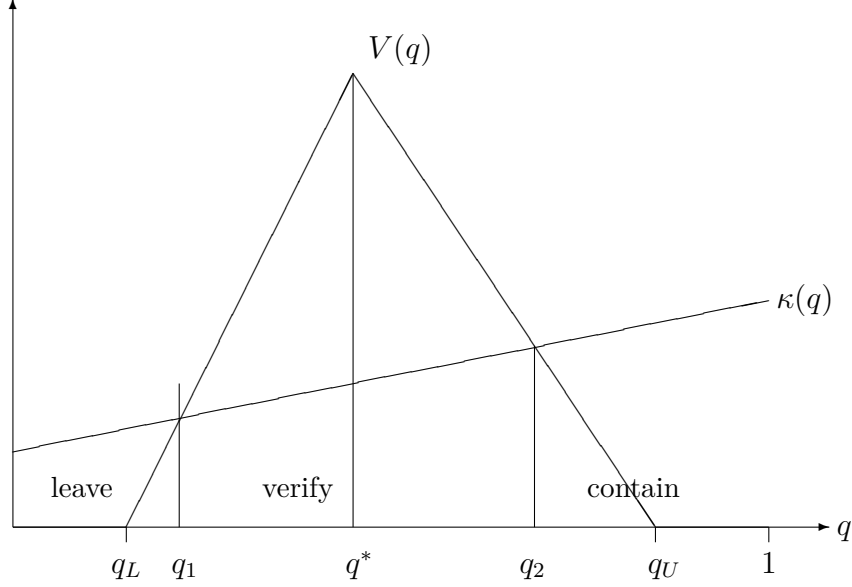


Figure 1: Schematic of Lemma 3 and Theorem 1 (not to scale; no parameter values). The value of information V is a tent on (q_L, q_U) with its peak at the passive threshold q^* ; the verification cost κ increases with the belief because harm during the delay accrues only under compromise. Verification pays on (q_1, q_2) , where the tent lies above the cost line. Raising the delay τ tilts the cost line upward and shrinks the interval (Theorem 2).

For an operator the rule reads as follows. Below q_1 the check costs more than it is expected to save, and the operator leaves. Above q_2 a compromise is likely enough that the operator contains without checking. In between, the operator verifies and follows the outcome. The verification interval always contains the passive threshold q^* when it is non-empty, and it lies inside the interval (q_L, q_U) on which the check has any value at all. Its two ends move inward as the check becomes more expensive or slower: c_v raises the lower end and lowers the upper one, and the delay term $h_K\tau$ enters both denominators.

4.5. The critical delay

Theorem 2 (Critical delay). Under Assumptions 1, 3 and 4, write $S(\tau)$ for the verification region at delay τ , all other parameters fixed. (i) *Nesting.* $\tau < \tau'$ implies $S(\tau') \subseteq S(\tau)$; if $h_K > 0$ and $S(\tau) \neq \emptyset$, then q_2 is strictly decreasing in τ , and q_1 is strictly increasing in τ unless $c_v = 0$ and

$t = 1$, in which case $q_1 = 0$. (ii) *Critical delay.* If $h_K > 0$ and $V(q^*) > c_v$, let

$$\tau^* = \frac{V(q^*) - c_v}{q^* h_K} = \frac{(s + t - 1)a - c_v/q^*}{h_K} > 0.$$

Then $S(\tau) \neq \emptyset$ if and only if $\tau < \tau^*$. (iii) *Degenerate cases.* If $V(q^*) \leq c_v$, then $S(\tau) = \emptyset$ for every $\tau \geq 0$. If $h_K = 0$, $S(\tau)$ does not depend on τ . (iv) *How the interval vanishes.* If $c_v > 0$ or $t < 1$, then $q_1 \rightarrow q^*$ and $q_2 \rightarrow q^*$ as $\tau \uparrow \tau^*$: the interval shrinks continuously to the point q^* . If $c_v = 0$ and $t = 1$, then $S(\tau) = (0, q_2(\tau))$ for $\tau < \tau^*$ with $q_2(\tau) \downarrow q^*$, and $S(\tau^*) = \emptyset$: the interval vanishes discontinuously.

Theorem 2 is an operational corollary of Lemma 3 and Theorem 1, not a new decision-theoretic result. Once the value of information is known to be a tent peaking at q^* , and the cost of the check is affine and increasing in the belief, the check pays at some belief if and only if it pays at q^* ; solving that inequality for the delay gives τ^* . Costs of delay that depend on the belief about the underlying state are known in time-critical decision analysis [3], and threshold analysis has been used to trade the delay of a more accurate test against its lost clinical benefit [4]. What Theorem 2 packages for attack-fault triage is the critical delay as a single closed-form quantity.

For an operator, τ^* is a time budget for the check. A verification procedure that takes longer than τ^* should never be run, at any belief: the operator should act on the passive rule instead. Below τ^* the verification interval shrinks as the check slows down, and the operator verifies over a narrower band of beliefs around q^* . The budget grows with the accuracy of the check (through $s + t - 1$) and shrinks with the harm rate h_K and with the fixed cost c_v . If the fixed cost alone exceeds the peak value of information, no delay is short enough. In the limiting case of a free check that never flags a fault ($c_v = 0$ and $t = 1$), the interval reaches down to zero belief and disappears all at once at τ^* ; otherwise it closes continuously onto q^* .

4.6. Comparison with the passive cost of ambiguity

The results above are stated for a belief. To compare verification with Paper 1's passive analysis, the decision problem is stated over policies that act on the order-flow observation.

Definition 4 (Decision problem with verification; verify-first cost of ambiguity). The cause M is K with prior probability $p = P(K) \in (0, 1)$ and F otherwise. Given M , the order-flow observation Y has law P_M^Y , and

the verification outcome Z has the law of Definition 2, jointly as in Assumption 2. Write $L(\text{contain}, K) = C_K$, $L(\text{contain}, F) = C_F$, $L(\text{leave}, K) = H_K$, $L(\text{leave}, F) = H_F$ (Definition 1). A *verify-first policy* is a quadruple of measurable maps $\delta, \pi_0, \pi_+, \pi_-$ from observations y to $[0, 1]$: with probability $\delta(y)$ the operator verifies and then contains with probability $\pi_Z(y)$; otherwise it contains with probability $\pi_0(y)$. Its expected cost is

$$J(\delta, \pi) = \mathbb{E} \left[(1 - \delta(Y)) (\pi_0(Y) L(\text{contain}, M) + (1 - \pi_0(Y)) L(\text{leave}, M)) \right. \\ \left. + \delta(Y) (c_v + \mathbb{I}[M = K] h_K \tau + \pi_Z(Y) L(\text{contain}, M) + (1 - \pi_Z(Y)) L(\text{leave}, M)) \right].$$

A *passive policy* is a verify-first policy with $\delta \equiv 0$. Let R_Y^* and $R_{Y,v}^*$ be the infima of J over passive and over verify-first policies, $R_{\text{oracle}} = p \min\{C_K, H_K\} + (1 - p) \min\{C_F, H_F\}$ the expected cost of an operator who observes M , and

$$\text{CoA}^* = R_Y^* - R_{\text{oracle}}, \quad \text{CoA}_v^* = R_{Y,v}^* - R_{\text{oracle}}$$

the passive and the verify-first cost of ambiguity.

Corollary 1 (Verification lowers the confounded cost of ambiguity). Under Assumptions 1–4, in the setting of Definition 4, suppose the order-flow laws coincide, $P_K^Y = P_F^Y$ (the confounded case). Then: (i) $\text{CoA}^* = \min\{p a, (1 - p) b\} > 0$; (ii) the verify-first cost of ambiguity is

$$\text{CoA}_v^* = \min\{\text{CoA}^*, E(p)\}, \quad E(p) = c_v + p h_K \tau + p(1 - s) a + (1 - p)(1 - t) b,$$

equivalently $\text{CoA}_v^* = \text{CoA}^* - (V(p) - \kappa(p))^+$, and it is attained by the policy of Theorem 1 applied at the belief $q = p$, whatever Y ; (iii) $\text{CoA}_v^* < \text{CoA}^*$ if and only if $p \in S$, that is, if and only if $V(q^*) > c_v + q^* h_K \tau$ and $q_1 < p < q_2$ (Theorem 1); if $h_K > 0$ and $V(q^*) > c_v$, this holds if and only if $\tau < \tau^*$ and $q_1(\tau) < p < q_2(\tau)$ (Theorem 2), and if $V(q^*) \leq c_v$ it holds for no p and no τ ; (iv) over $p \in (0, 1)$, the reduction $\text{CoA}^* - \text{CoA}_v^*$ is largest at $p = q^*$, where it equals $(V(q^*) - c_v - q^* h_K \tau)^+$; (v) $\text{CoA}_v^* = 0$ if and only if $s = t = 1$, $c_v = 0$ and $h_K \tau = 0$. Under the correspondence of Paper 1, Definition D8 ($H_F = 0$), CoA^* in (i) is the cost of ambiguity of Paper 1, Proposition 1, so (iii) states when verification brings the cost of ambiguity strictly below that value.

In the confounded case the order flow carries no information about the cause, and the passive cost of ambiguity in part (i) is the value that Paper 1 shows no passive policy can avoid. Corollary 1 states when verification brings it strictly lower: exactly when the prior lies in the verification region of

Theorem 1, so, for a harmful compromise and a check whose fixed cost is below the peak value of information, only when the check is faster than τ^* . The reduction is largest when the prior sits at the passive threshold, and the cost of ambiguity is removed entirely only by a perfect and free check that accrues no harm during its delay. The optimal verify-first policy ignores the order flow, which here carries no information, and applies the rule of Theorem 1 to the prior.

4.7. Degenerate parameters

Remark 1 (Degenerate parameters). (a) If $s + t = 1$, then $q^+ = q^- = q$ whenever defined, $V \equiv 0$, and $S = \emptyset$ for all $c_v \geq 0$, $\tau \geq 0$. (b) If Assumption 1 fails with $a \leq 0 \leq b$ or $b \leq 0 \leq a$ (so $q^* \notin (0, 1)$ or is undefined), one terminal action is weakly optimal at every belief, R is affine, $V \equiv 0$, and $S = \emptyset$. (c) The requirement $c_v \geq 0$ in Assumption 3 is needed for Theorem 1: if $c_v < 0$, S can contain beliefs near 0 where $V = 0$ and can then be disconnected.

Parts (a) and (b) say that the results above need a check that is informative and a decision that is not settled in advance. Part (c) explains why Assumption 3 requires a non-negative cost: a check that pays the operator to run it would be worth running at beliefs where its outcome cannot change the action, and the verification region could then fail to be an interval.

4.8. Proofs

Proof of Lemma 1. Let $L_c(q) = qC_K + (1 - q)C_F$ (contain) and $L_\ell(q) = qH_K + (1 - q)H_F$ (leave). Their difference is affine:

$$D(q) := L_\ell(q) - L_c(q) = q(H_K - C_K) - (1 - q)(C_F - H_F) = qa - (1 - q)b = (a + b)(q - q^*),$$

with $q^* = b/(a + b)$. By Assumption 1, $a, b > 0$, so $a + b > 0$ and $q^* \in (0, 1)$. Hence $D < 0$ (leave strictly cheaper) for $q < q^*$ and $D > 0$ (contain strictly cheaper) for $q > q^*$. Since $\min\{L_\ell, L_c\} = L_\ell - \max\{0, D\}$,

$$R(q) = L_\ell(q) - (a + b)(q - q^*)^+.$$

R is the minimum of two affine functions, hence continuous and concave; it is piecewise linear with slopes $H_K - H_F$ on $[0, q^*]$ and $C_K - C_F$ on $[q^*, 1]$, which differ by $-(a + b) \neq 0$, so there is exactly one kink, at q^* . $\square$

Proof of Lemma 2. Fix y with $P(Y = y) > 0$ (for densities, replace probabilities of Y by densities; the argument is unchanged), write $q = q(y)$, and let z be an outcome with $P(Z = z | Y = y) > 0$. By Bayes' rule and Assumption 2,

$$P(K | y, z) = \frac{P(z | K) P(y | K) P(K)}{\sum_{c \in \{K, F\}} P(z | c) P(y | c) P(c)}.$$

Divide numerator and denominator by $P(y) = \sum_c P(y | c) P(c)$ and use $P(y | c) P(c) / P(y) = P(c | y)$:

$$P(K | y, z) = \frac{P(z | K) q}{P(z | K) q + P(z | F) (1 - q)}.$$

For $z = +$ this is $qs / (qs + (1 - q)(1 - t)) = q^+$; for $z = -$ it is $q(1 - s) / (q(1 - s) + (1 - q)t) = q^-$. Likewise $P(Z = z | Y = y) = P(z | K) q + P(z | F) (1 - q)$, which is $P(+)$ or $P(-)$ of Definition 2 at $q = q(y)$. So, conditionally on $Y = y$, verification acts on the belief $q(y)$ exactly as in Definition 2 with the same (s, t) , and every quantity built from Definitions 2–3 (R_v , V , S) is evaluated at $q(y)$. Nothing in the argument uses the law of Y beyond $q(y)$; in particular it holds when $P(Y | K) = P(Y | F)$, in which case $q(y)$ equals the prior. Without Assumption 2 the factorisation in the first display fails and the update would depend on y through $P(z | y, c)$. $\square$

Proof of Lemma 3. Step 1 (homogeneous form). For an outcome z write $\alpha_z = P(z | K) q$ and $\beta_z = P(z | F) (1 - q)$, so $P(z) = \alpha_z + \beta_z$ and, by Definition 3, $P(z)R(q^z) = \min\{\alpha_z C_K + \beta_z C_F, \alpha_z H_K + \beta_z H_F\}$. The difference of the two arguments is $(\alpha_z H_K + \beta_z H_F) - (\alpha_z C_K + \beta_z C_F) = \alpha_z a - \beta_z b$, so

$$P(z)R(q^z) = (\alpha_z H_K + \beta_z H_F) - \max\{0, \alpha_z a - \beta_z b\}.$$

Here $\alpha_+ a - \beta_+ b = qsa - (1 - q)(1 - t)b = G^+(q)$ and $\alpha_- a - \beta_- b = q(1 - s)a - (1 - q)tb = -G^-(q)$. Summing over z and using $\alpha_+ + \alpha_- = q$, $\beta_+ + \beta_- = 1 - q$, with $L_\ell(q) = qH_K + (1 - q)H_F$:

$$P(+)R(q^+) + P(-)R(q^-) = L_\ell(q) - \max\{0, G^+(q)\} - \max\{0, -G^-(q)\}.$$

By the proof of Lemma 1, $R(q) = L_\ell(q) - \max\{0, D(q)\}$ with $D(q) = qa - (1 - q)b$, and summing the two outcome terms gives $G^+(q) - G^-(q) = D(q)$. Therefore

$$V(q) = \max\{0, G^+\} + \max\{0, -G^-\} - \max\{0, G^+ - G^-\}. \quad (1)$$

Step 2 (zeros and the value at q^).* G^+ is affine with slope $sa + (1-t)b > 0$ (as $a, b > 0$ and $s > 1-t \geq 0$); G^- is affine with slope $-(tb + (1-s)a) < 0$ (as $t > 1-s \geq 0$). Their unique zeros are q_L and q_U of the statement, and the formulas give $0 \leq q_L$ and $q_U \leq 1$. At q^* , $D(q^*) = 0$, so $G^+(q^*) = G^-(q^*) = q^*sa - (1-q^*)(1-t)b$. Since $q^*a = (1-q^*)b = ab/(a+b)$, this equals $(s+t-1)ab/(a+b)$, which is > 0 by Assumption 3. As G^+ is increasing with $G^+(q^*) > 0$, $q_L < q^*$; as G^- is decreasing with $G^-(q^*) > 0$, $q^* < q_U$. This proves (i).

Step 3 (the tent). Evaluate (1) on three ranges.

- $q \leq q_L$: $G^+ \leq 0$ and (as $q < q_U$) $G^- > 0$, so $G^+ - G^- < 0$; every term of (1) vanishes and $V = 0$.
- $q \geq q_U$: $G^- \leq 0$ and (as $q > q_L$) $G^+ > 0$, so $G^+ - G^- > 0$ and $V = G^+ - G^- - (G^+ - G^-) = 0$.
- $q_L < q < q_U$: $G^+ > 0$ and $G^- > 0$, so $V = G^+ - \max\{0, G^+ - G^-\} = \min\{G^+, G^-\} > 0$.

Hence $V = \max\{0, \min\{G^+, G^-\}\}$ on $[0, 1]$, and $V > 0$ exactly on (q_L, q_U) . On that interval, $G^+ < G^-$ iff $D < 0$ iff $q < q^*$. So $V = G^+$ (affine, strictly increasing) on $[q_L, q^*]$ and $V = G^-$ (affine, strictly decreasing) on $[q^*, q_U]$; V is continuous, and its unique maximiser is q^* , with $V(q^*) = (s+t-1)ab/(a+b) = (s+t-1)aq^*$. This proves (iii) and (iv).

Step 4 (posterior form of (ii)). For $q \in (0, 1)$, $P(+)>0$ and $P(-)>0$ because $s > 0$ and $t > 0$. Using $(a+b)q^* = b$,

$$(a+b)P(+)(q^+ - q^*) = (a+b)qs - bP(+) = qsa - (1-q)(1-t)b = G^+(q),$$

and in the same way $(a+b)P(-)(q^* - q^-) = G^-(q)$. So $G^+(q) > 0 \iff q^+ > q^*$ and $G^-(q) > 0 \iff q^- < q^*$; hence $q \in (q_L, q_U) \iff q^- < q^* < q^+$. At $q \in \{0, 1\}$, $V = 0$, and every defined posterior equals q , so the posteriors cannot straddle $q^* \in (0, 1)$: both sides of (ii) are false. $\square$

Proof of Theorem 1. Assumption 4 is what makes Definition 3 the correct verify-first risk: under K (probability q) verification adds $c_v + h_K\tau$ to the terminal cost, under F it adds c_v , the terminal costs and (s, t) are unchanged by the delay, and the terminal action is chosen optimally on the posterior, giving $c_v + qh_K\tau + \sum_z P(z)R(q^z)$. So $R_v < R$ iff $V > \kappa$, with $\kappa(q) = c_v + qh_K\tau$ affine and nondecreasing.

Step 1 ($S \subseteq (q_L, q_U)$). By Assumption 3, $\kappa(q) \geq 0$ on $[0, 1]$. By Lemma 3, $V = 0$ outside (q_L, q_U) , so $V > \kappa$ fails there.

Step 2 (S is an interval). Let $\varphi(q) = \min\{G^+(q), G^-(q)\} - \kappa(q)$ on $[0, 1]$. It is the minimum of two affine functions minus an affine function, hence concave and continuous. On (q_L, q_U) , $V = \min\{G^+, G^-\}$ (Lemma 3), and outside it $\min\{G^+, G^-\} \leq 0$ (since $G^+ \leq 0$ for $q \leq q_L$ and $G^- \leq 0$ for $q \geq q_U$), so $\varphi \leq -\kappa \leq 0$ there. Hence $S = \{q : \varphi(q) > 0\}$, a strict superlevel set of a continuous concave function: an open (relative to $[0, 1]$) interval, possibly empty. Since $\varphi(0) = \min\{-(1-t)b, tb\} - c_v \leq 0$ and $\varphi(1) = \min\{sa, -(1-s)a\} - c_v - h_K\tau \leq 0$, the endpoints are excluded and S is an open interval in $(0, 1)$.

Step 3 (non-emptiness criterion). On $[q^*, q_U]$, $\varphi = G^- - \kappa$ is strictly decreasing (G^- strictly decreasing, κ nondecreasing). On $[q_L, q^*]$, $\varphi = G^+ - \kappa$ is affine with slope $\sigma = sa + (1-t)b - h_K\tau$. If $\sigma \leq 0$, φ is nonincreasing on $[q_L, q_U]$, so $\sup_{(q_L, q_U)} \varphi \leq \varphi(q_L) = -\kappa(q_L) \leq 0$ and $S = \emptyset$. If $\sigma > 0$, φ is strictly increasing then strictly decreasing, so $\max_{[q_L, q_U]} \varphi = \varphi(q^*) = V(q^*) - \kappa(q^*)$. Either way, $S \neq \emptyset$ implies $\varphi(q^*) > 0$ (in the first case S is empty; in the second the maximum is at q^*). Conversely $\varphi(q^*) > 0$ gives $q^* \in S$. Hence $S \neq \emptyset \iff V(q^*) > c_v + q^*h_K\tau$, and then $q^* \in S$. Also, $S \neq \emptyset$ forces $\sigma > 0$: if $\sigma \leq 0$ we showed $S = \emptyset$.

Step 4 (thresholds). Assume $S \neq \emptyset$. Then $\sigma > 0$, and the left endpoint q_1 of S is the zero of the affine function $G^+ - \kappa = \sigma q - ((1-t)b + c_v)$ on $[q_L, q^*]$ if $\varphi(q_L) < 0$, or q_L itself if $\varphi(q_L) = 0$; in both cases $q_1 = (c_v + (1-t)b)/\sigma$ (when $\varphi(q_L) = 0$, $\kappa(q_L) = 0$, so $c_v = 0$ and $q_L h_K\tau = 0$, and the formula returns q_L). The right endpoint q_2 is the zero of $G^- - \kappa = tb - c_v - q(tb + (1-s)a + h_K\tau)$ on $(q^*, q_U]$, i.e. $q_2 = (tb - c_v)/(tb + (1-s)a + h_K\tau)$; its denominator is $\geq tb > 0$. As $G^+(q_1) = \kappa(q_1) \geq 0$ we get $q_1 \geq q_L$, and as $G^-(q_2) = \kappa(q_2) \geq 0$ we get $q_2 \leq q_U$; $q_1 < q^* < q_2$ because $q^* \in S$ and S is open.

Step 5 (policy). For $q \notin S$ the optimal action is the passive one of Lemma 1; since $q_1 < q^* < q_2$, this is leave on $q \leq q_1$ and contain on $q \geq q_2$. For $q \in S \subseteq (q_L, q_U)$, Lemma 3(ii) gives $q^- < q^* < q^+$, so by Lemma 1 the optimal terminal action after verification is contain on $Z = +$ and leave on $Z = -$.

Step 6 ($c_v = 0, h_K\tau = 0$). Then $\kappa \equiv 0$ and $S = \{V > 0\} = (q_L, q_U)$ by Lemma 3(ii); the threshold formulas reduce to q_L and q_U . $\square$

Proof of Theorem 2. Throughout, only τ varies; V, q^*, q_L, q_U do not depend on τ (Assumption 4: s, t and the loss table are delay-free).

(i) *Nesting.* For $q \in [0, 1]$, $\kappa_\tau(q) = c_v + qh_K\tau$ is nondecreasing in τ because $qh_K \geq 0$. So $V(q) > \kappa_{\tau'}(q)$ implies $V(q) > \kappa_\tau(q)$ for $\tau < \tau'$, i.e. $S(\tau') \subseteq S(\tau)$. If $h_K > 0$ and $S(\tau) \neq \emptyset$, then $V(q^*) > c_v$, so by (ii) below $S \neq \emptyset$ on all of $[0, \tau^*)$ and the formulas of Theorem 1 apply there. $q_2 = (tb - c_v)/(tb + (1 - s)a + h_K\tau)$ has a positive numerator (because $q_2 > q^* > 0$ and the denominator is positive) and a denominator strictly increasing in τ , so it is strictly decreasing. $q_1 = (c_v + (1 - t)b)/(sa + (1 - t)b - h_K\tau)$ has a positive denominator strictly decreasing in τ and a nonnegative numerator, zero iff $c_v = 0$ and $t = 1$; so q_1 is strictly increasing unless $c_v = 0$ and $t = 1$, where $q_1 = 0$.

(ii) *Critical delay.* By Theorem 1, $S(\tau) \neq \emptyset \iff V(q^*) > c_v + q^*h_K\tau$. With $h_K > 0$ and $q^* > 0$, this is $\tau < (V(q^*) - c_v)/(q^*h_K) = \tau^*$, which is > 0 when $V(q^*) > c_v$. By Lemma 3(iv), $V(q^*)/q^* = (s + t - 1)a$, giving the second expression for τ^* .

(iii) *Degenerate cases.* If $V(q^*) \leq c_v$, then $V(q^*) \leq c_v + q^*h_K\tau$ for every $\tau \geq 0$, so $S(\tau) = \emptyset$ by Theorem 1. If $h_K = 0$, κ does not involve τ , so neither does S .

(iv) *How the interval vanishes.* Let $h_K > 0$, $V(q^*) > c_v$, and $\tau \uparrow \tau^*$. The denominator of q_2 is continuous and positive at τ^* , so $q_2(\tau) \rightarrow q_2(\tau^*)$, the zero of $G^- - \kappa_{\tau^*}$. At τ^* , $G^-(q^*) = V(q^*) = \kappa_{\tau^*}(q^*)$, so that zero is q^* (it is unique: $G^- - \kappa_{\tau^*}$ is strictly decreasing). For q_1 : at τ^* the left-piece function $G^+ - \kappa_{\tau^*}$ vanishes at q^* and equals $-\kappa_{\tau^*}(q_L) = -(c_v + q_L h_K \tau^*)$ at q_L . If $c_v > 0$ or $t < 1$ (the latter giving $q_L > 0$), this is < 0 , so the slope $\sigma(\tau^*) = sa + (1 - t)b - h_K\tau^*$ is > 0 , the formula for q_1 is continuous at τ^* , and $q_1(\tau) \rightarrow$ the zero of $G^+ - \kappa_{\tau^*}$, which is q^* . If $c_v = 0$ and $t = 1$, then $q_L = 0$, $G^+(q) = qsa$ and $\kappa_\tau(q) = qh_K\tau$, and $\tau^* = sa/h_K$ (as $s + t - 1 = s$). For $\tau < \tau^*$, $G^+ - \kappa_\tau = q(sa - h_K\tau) > 0$ on $(0, q^*]$, so $S(\tau) = (0, q_2(\tau))$ with $q_2(\tau) \downarrow q^*$ by (i) and the limit above; at τ^* , $G^+ - \kappa_{\tau^*} \equiv 0$ on $[0, q^*]$ and $S(\tau^*) = \emptyset$ by (ii). So the interval drops from $(0, q^*)$ (in the limit) to $\emptyset$. $\square$

Proof of Corollary 1. Reduction to a constant belief. By Assumption 2 the joint law of (M, Y, Z) is $P(M = m) P_m^Y(dy) P(Z = z | m)$. Let $\mu = P_K^Y = P_F^Y$. Then Y has law μ and is independent of (M, Z) , and $P(M = m, Z = z) = P(m) P(z | m)$. Write $P(z, K) = P(z | K)p$ and $P(z, F) = P(z | F)(1 - p)$ as in Definition 3. Conditioning on $Y = y$ in Definition 4 gives

$J(\delta, \pi) = \int \ell(y) \mu(dy)$ with

$$\begin{aligned} \ell(y) = & (1 - \delta(y)) \left[\pi_0(y) (pC_K + (1-p)C_F) + (1 - \pi_0(y)) (pH_K + (1-p)H_F) \right] \\ & + \delta(y) \left[c_v + p h_K \tau + \sum_{z \in \{+, -\}} \left(\pi_z(y) (P(z, K)C_K + P(z, F)C_F) + (1 - \pi_z(y)) (P(z, K)H_K + P(z, F)H_F) \right) \right] \end{aligned}$$

Here c_v and $h_K \tau$ are paid only on verification, the latter only under K (Assumption 4). Each expression of the form $\pi x + (1 - \pi) x'$ with $\pi \in [0, 1]$ is at least $\min\{x, x'\}$. So the first bracket is at least $R(p)$ (Definition 1). The second is at least $c_v + p h_K \tau + \sum_z \min\{P(z, K)C_K + P(z, F)C_F, P(z, K)H_K + P(z, F)H_F\} = R_v(p)$, by the reading of Definition 3. Hence $\ell(y) \geq (1 - \delta(y))R(p) + \delta(y)R_v(p) \geq \min\{R(p), R_v(p)\}$ for every y , and $J \geq \min\{R(p), R_v(p)\}$. The constant policy that verifies iff $R_v(p) < R(p)$ and takes the minimizing terminal action in each case attains equality. So

$$R_{Y,v}^* = \min\{R(p), R_v(p)\}, \quad R_Y^* = R(p)$$

(the second with $\delta \equiv 0$). The attaining policy depends on p only, not on y . When $p \in S$ it is the verify branch of Theorem 1 at $q = p$: contain on $Z = +$ and leave on $Z = -$, because $p^- < q^* < p^+$ by Lemma 3(ii). Otherwise it is the passive rule of Lemma 1.

(i) By Assumption 1, $\min\{C_K, H_K\} = C_K$ and $\min\{C_F, H_F\} = H_F$, so $R_{\text{oracle}} = pC_K + (1-p)H_F$. Subtracting it from both terms of $R(p)$ gives $\text{CoA}^* = \min\{(1-p)(C_F - H_F), p(H_K - C_K)\} = \min\{pa, (1-p)b\}$. This is > 0 since $a, b > 0$ and $0 < p < 1$.

(ii) By Definition 3, $R_v(p) = \kappa(p) + R(p) - V(p)$. So $\text{CoA}_v^* = \min\{R(p), R_v(p)\} - R_{\text{oracle}} = \text{CoA}^* - (V(p) - \kappa(p))^+$. With $G^\pm$ as in Lemma 3,

$$\text{CoA}^* - E(p) = \min\{pa - p(1-s)a - (1-p)(1-t)b, (1-p)b - p(1-s)a - (1-p)(1-t)b\} - \kappa(p) = \min\{G^+, G^-\}$$

By Lemma 3, $V = \max\{0, \min\{G^+, G^-\}\}$, and $\kappa(p) \geq 0$ by Assumption 3. Suppose $\min\{G^+, G^-\} > \kappa$. Then $V = \min\{G^+, G^-\}$ and $(V - \kappa)^+ = \min\{G^+, G^-\} - \kappa$. Otherwise $V \leq \max\{0, \kappa\} = \kappa$, and $(V - \kappa)^+ = 0 = (\min\{G^+, G^-\} - \kappa)^+$. In both cases $(V - \kappa)^+ = (\text{CoA}^* - E)^+$, so $\text{CoA}_v^* = \text{CoA}^* - (\text{CoA}^* - E)^+ = \min\{\text{CoA}^*, E(p)\}$. (Direct check: $E(p)$ is the cost of the verify branch above minus R_{oracle} . That is c_v , plus $h_K \tau$ under K , a miss a with probability $1 - s$ under K , and a false containment b with probability $1 - t$ under F .)

(iii) By (ii), $\text{CoA}_v^* < \text{CoA}^* \iff V(p) > \kappa(p) \iff p \in S$ (Definition 3). By Theorem 1, $S \neq \emptyset \iff V(q^*) > c_v + q^* h_K \tau$, and then $S = (q_1, q_2)$. If $h_K > 0$ and $V(q^*) > c_v$, Theorem 2(ii) turns the non-emptiness condition into $\tau < \tau^*$. If $V(q^*) \leq c_v$, then $S(\tau) = \emptyset$ for all $\tau \geq 0$ by Theorem 2(iii).

(iv) The reduction is $D(p) = (V(p) - \kappa(p))^+$, and $V - \kappa \leq -\kappa \leq 0$ outside (q_L, q_U) . On $[q^*, q_U]$, $V - \kappa = G^- - \kappa$ is strictly decreasing (Lemma 3(iii); κ is nondecreasing). On $[q_L, q^*]$, $V - \kappa$ is affine. It is ≤ 0 at q_L and equals $V(q^*) - \kappa(q^*)$ at q^* . So its maximum over $[q_L, q^*]$ is attained at an endpoint, and $\max_p D(p) = (V(q^*) - \kappa(q^*))^+ = (V(q^*) - c_v - q^* h_K \tau)^+$, attained at q^* . When this maximum is positive, D is strictly increasing on $[q_L, q^*]$ (an affine function going from ≤ 0 to a positive value) wherever it is positive, and strictly decreasing on $[q^*, q_U]$. So the maximizer is unique.

(v) Since $\text{CoA}^* > 0$ by (i), $\text{CoA}_v^* = 0$ iff $E(p) = 0$. Each of the four terms of $E(p)$ is nonnegative (Assumption 3). With $p \in (0, 1)$ and $a, b > 0$, all four vanish iff $c_v = 0$, $h_K \tau = 0$, $s = 1$ and $t = 1$. $\square$

Proof of Remark 1. Step 1 of the proof of Lemma 3 used only Definitions 1–3 and holds without Assumptions 1 and 3: $V = \max\{0, G^+\} + \max\{0, -G^-\} - \max\{0, G^+ - G^-\}$, with $G^+ - G^- = D = qa - (1 - q)b$.

(a) If $s + t = 1$, then $P(z | K) = P(z | F)$ for both outcomes, so $q^z = q$ whenever $P(z) > 0$. Then $\sum_z P(z) R(q^z) = R(q)$ (the $P(z) = 0$ term is zero by Definition 3), so $V \equiv 0$. As $\kappa \geq 0$, $V > \kappa$ never holds and $S = \emptyset$.

(b) If $a \leq 0 \leq b$, then $D(q) = qa - (1 - q)b \leq 0$ on $[0, 1]$, so leaving is weakly optimal everywhere and $R = L_\ell$ is affine. Each outcome term satisfies $\alpha_z a - \beta_z b \leq 0$ (as $\alpha_z, \beta_z \geq 0$), so $\sum_z P(z) R(q^z) = \sum_z (\alpha_z H_K + \beta_z H_F) = L_\ell(q) = R(q)$ and $V \equiv 0$. The case $b \leq 0 \leq a$ is symmetric ($D \geq 0$, contain weakly optimal everywhere, each $\alpha_z a - \beta_z b \geq 0$, so $\sum_z P(z) R(q^z) = L_c(q) = R(q)$). In both, $S = \emptyset$ since $\kappa \geq 0$.

(c) Take $c_v < 0$, $t < 1$ (so $q_L > 0$ under Assumption 1), $h_K > 0$, and $\tau > 0$ small enough that $V(q^*) > q^* h_K \tau$. Choose $c_v \in (-q_L h_K \tau, 0)$. Then $q^* \in S$, because $V(q^*) > q^* h_K \tau > \kappa(q^*)$. At $q = 0$, $V(0) = 0 > c_v = \kappa(0)$, so $0 \in S$. At $q = q_L$, $V(q_L) = 0 < c_v + q_L h_K \tau = \kappa(q_L)$, so $q_L \notin S$. Since $0 < q_L < q^*$, S is not an interval. $\square$

5. Empirical evaluation on Paper 1 predictions

Planned. Era-3 classifier posteriors (prior-shift corrected) as passive beliefs; lead with the low-prior cells where passive classification fails; report FPR/FNR beside cost.

6. Discussion

Planned.

7. Conclusion

Planned.

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